Additive Baselines on the 025 Build

The trigenic ladder under a random and a query-pair-disjoint split, in preparation for the main text

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September 12, 2026

Abstract

Five nulls that cannot represent a three-way interaction, and a nonlinear baseline on the same features that can, are fit on exactly the 376,732 trigenic records and the two splits the 025 transformer arms train on. Under the published random-over-records split the per-gene additive ridge reaches 0.400 test Pearson, a model that sees only which Kuzmin query double a triple carries reaches 0.390, and the transformer checkpoints reach 0.438 to 0.455. Holding out whole query doubles collapses the additive ridge to 0.185 on the one disjoint split the transformer trains on, and to 0.127 ± 0.033 over five disjoint folds, and drops the nonlinear baseline below the additive one. Four transformer runs have trained on the disjoint split, one in the 010 configuration and three with a fixed sequence embedding and a constant learning rate. Every one of them peaks on validation between epoch 2 and epoch 7, at 0.199 to 0.270 against an additive validation null of 0.150, and every one falls back afterward; the 010 configuration ends below the null at 0.131. None has a test score, so whether the transformer's margin over additivity survives held-out screens is not yet measured, and the measurements that would settle it are listed.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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Contents

1 The question, and the state of the evidence ✗

A model that claims to learn genetic interaction has to beat a null that cannot represent interaction, on data where the null cannot borrow the answer from the training set. The 010 trigenic model was never measured against such a null, and its published split lets every model score the identity of the screen a record came from. Both gaps are closed here on the 025 build, which holds the 010 records with bit-identical labels, under two splits: the published random split (arm R) and a split that never shares a Kuzmin query double between parts (arm Q).

Measured. Five nulls and the nonlinear baseline on both arms, on test (Table 4, Figure 1a,b). Arm R reproduces the 010 numbers. On arm Q the additive ridge falls from 0.400 to 0.185, the screen-only model to zero, and the nonlinear baseline from 0.432 to 0.141, below the ridge. Four transformer runs on arm Q, every one peaking on validation by epoch 7 and declining afterward (Figure 2a,b).
Not measured. Any transformer test score on arm Q, and any replicate of the 010 configuration on arm Q. Figure 2c is the placeholder for both.

Vocabulary. **Arm R** and **arm Q** are the two training arms of experiment 025, one per split; every baseline is fit under both. The **ladder** is the ordered set of baselines B0 through B5 and the transformer, read left to right by interaction capacity. A **query pair** is the double mutant a Kuzmin trigenic screen was built around; every record in the build is one query pair plus one **array gene**, and 420 query pairs recur across the 376,732 records. A **screen** is the set of records sharing a query pair. A **validation maximum** is the largest validation Pearson over a run's logged epochs; it is an upward-biased order statistic, and it is never set beside a test number without that label.

2 Data and the two splits ✗

The label. Write f_i, f_{ij} and f_{ijk} for the fitness of the single, double and triple mutant relative to wild type, and $\varepsilon_{ij} = f_{ij} - f_i f_j$ for the digenic interaction. The trigenic interaction score is

$$\tau_{ijk} \;=\; f_{ijk} - f_i f_j f_k - f_i\,\varepsilon_{jk} - f_j\,\varepsilon_{ik} - f_k\,\varepsilon_{ij}, \tag{1}$$

▲ [external: Kuzmin et al. 2018, `10.1126/science.aao1729`; the definition of the released adjusted score, not re-derived here] a third-order residual: the multiplicative part of fitness and every pairwise term are subtracted before the number is reported, so τ is by construction the part of the triple phenotype that no single or double mutant predicts. The loaders store it as `gene_interaction`.

The records. Subset S0 of the 025 build is its 376,732 trigenic records over 4,352 genes. Matching them to the 010 build by genotype recovers every one; 376,526 labels are bit-identical and the remaining 206 differ by at most 5.6×10^{-17} (`experiments/025-solid-growth/scripts/label_parity_010_vs_025.py`). Every model below is fit on these records and nothing else.

The screen structure. The records contain 739,315 distinct unordered gene pairs, of which 420 occur five or more times. Those 420 account for 376,733 pair instances, one more than the record count: every record contains exactly one recurring pair, and a single record contains two. The 420 pairs are the Kuzmin query doubles. Grouping records by their recurring pair recovers the screen exactly, and any model with a per-pair offset can absorb whatever is common to a screen.

The two arms. Arm R is 010's own split, random over records, carried into the 025 index space by genotype identity. Arm Q assigns whole query pairs to parts, so no screen appears in more than one part. The two arms hold out nearly the same number of records and differ only in what is held out.

Table 1: The two arms. Both partition the same 376,732 records. Each part of arm R carries every one of the 420 query pairs; the parts of arm Q carry disjoint sets of them. The split artifacts are the files the trainer loads, under `experiments/025-solid-growth/results/`.

	Arm R	Arm Q
Held-out unit	record	query pair
Recurring query pairs	420, shared by every part	420, no pair in two parts
Query pairs, train / val / test		331 / 43 / 46
Records, train / val / test	301,386 / 37,673 / 37,673	301,236 / 37,705 / 37,791
Split artifact	`pinned_splits_from_010_seed_42.json.gz`	`query_pair_disjoint_splits_025.json.gz`
Transformer run	GH 1598	GH 1640, and Table 5

What arm Q holds out. The query pair, not the genes. Every array gene of the validation and test parts occurs in training, as do 78 of the 86 validation and 88 of the 92 test query-pair genes, so 81 percent of validation records and 91 percent of test records have all three genes in the training set (Table 2). A model on arm Q is therefore asked to predict an unseen pair of mostly seen genes with a seen third gene, which is the generalization that scoring an unmeasured triple requires. The test part spreads its records over 1,182 distinct array genes against the validation part's 4,003; the two parts are different kinds of screen sample, and no analysis here uses that difference.

Table 2: What the held-out parts of arm Q share with its training part. A query pair is the most frequent recurring pair a record carries, the rule that built the split; the array gene is the record's third gene.

	Validation	Test
Records	37,705	37,791
Query pairs held out	43	46
Query pairs also in training	0	0
Distinct array genes	4,003	1,182
Array genes present in training	4003 of 4003	1182 of 1182
Query-pair genes present in training	78 of 86	88 of 92
Records with all three genes in training	81.3%	91.4%

3 The models

Let record n have perturbed set S_n of three genes, encoded as $x_n \in \{0,1\}^{|G|}$ over the $|G| = 4{,}352$ genes, with label $y_n = \tau$. A model is a set function $f : 2^G \to \mathbb{R}$, and its capacity to represent interaction is a structural property of the hypothesis class. For distinct genes g, h, k and any S disjoint from them,

$$\Delta_{gh} f(S) = f(S \cup \{g,h\}) - f(S \cup \{g\}) - f(S \cup \{h\}) + f(S), \qquad \Delta_{ghk} f(S) = \sum_{U \subseteq \{g,h,k\}} (-1)^{3-|U|} f(S \cup U). \qquad (2)$$

A model with $\Delta_{ghk} f \equiv 0$ cannot express a three-way interaction and is a null with respect to the label, whatever it is fit to.

B0, train mean. $\hat{y} = \bar{y}_{\text{train}}$. The zero point.

B1, additive per-gene ridge. $\hat{y}(S) = \beta_0 + \sum_{g \in S} \beta_g$, with β the ridge minimizer $(X^\top X + \alpha I)^{-1} X^\top \tilde{y}$ on the centered training labels and α chosen on validation over $\{10^{-2}, \ldots, 10^3\}$. Both finite differences vanish identically. B1 is the additive null of Visani, Verma and DeWitt (bioRxiv `2026.04.23.719915`) transplanted from protein variants to gene deletions.

B2, additive plus recurring pairs. B1 with one coefficient γ_{gh} per query pair in the training split. Each γ_{gh} is the mean additive residual over that screen, shrunk by $\alpha/(n_{gh} + \alpha)$ with n_{gh} in the hundreds to thousands, so B2 is B1 plus one nearly unshrunk offset per screen. Second-order capacity on observed pairs, no third-order capacity.

B3, screen mean. No fitted parameters. The mean training label over the records carrying the record's query pair; if no pair in the record recurs in training, the mean of its three gene means; failing that, the train mean. On arm R the first case always fires. On arm Q it never does, and B3 is a mean of gene means.

B4, query pair only. B2 without the gene terms: a per-screen offset and nothing else. The array gene contributes nothing, so B4's score is an upper bound on how much of the metric is the identity of the screen. On arm Q no test record carries a pair with a coefficient, and B4 is structurally zero.

B5, nonlinearity on the same features. A free 128-vector per gene, summed over the three perturbed genes and read out by a two-hidden-layer network, trained by gradient descent with early stopping on validation, three seeds. Exactly B1's feature space, but $\Delta_{ghk} \not\equiv 0$, so B5 is not a null. It is the smallest departure from B1 that has interaction capacity, and it isolates what capacity alone buys.

The transformer. The cell graph transformer of experiment 010, run on the 025 build with no change to architecture, loss, optimizer or schedule (`cgt_s0_r_kl_000` for arm R, `cgt_s0_q_kl_004` for arm Q). Its strain-dependent computation is a set function over the three perturbed gene identities, the same input the nulls see. The nine interaction graphs enter only through a penalty on layer-1 attention during training, so they shape the weights and are absent from the prediction function. The reading of the code that establishes both properties is in the 010 report and is not repeated.

Comparability. Every model is fit on the training part of the arm it is reported under, makes its free choices on the validation part, and is reported on test. The transformer's label normalizer is fit on the training part for arm Q and on all of S0 for arm R; Pearson is invariant to that choice, and the nulls use the training mean only.

Table 3: What each model can represent. The last two columns are properties of the hypothesis class, not fitted results. Only B5 and the transformer can express a three-way interaction, which is what the label is.

Model	Per-gene quantity	Set aggregation	Fit by	$\Delta_{gh} \not\equiv 0$	$\Delta_{ghk} \not\equiv 0$
B0	none	none	train mean	no	no
B1	free scalar	sum	ridge	no	no
B2	free scalar	sum, plus one offset per recurring pair	ridge	yes	no
B3	label mean over training triples	mean over pairs, else genes	label means	yes	no
B4	none	one offset per recurring pair	ridge	yes	no
B5	free 128-vector	sum, then MLP	gradient descent	yes	yes
CGT	free 180-vector through an encoder	attention over S, mean, MLP	gradient descent	yes	yes

4 The ladder under both splits ✗

Table 4: Test Pearson for every model under both arms, beside the 010 numbers. B0 through B4 are closed-form fits on a fixed split and carry no spread; B5 is the mean and standard deviation over three seeds. The three transformer rows stand on different footings, named in the row label: the 010 checkpoints were scored on the 010 test split of 37,673 records; GH 1598 and GH 1640 are validation maxima over 36 logged epochs, and neither has a test score.

	Arm R, random over records		Arm Q, query-pair disjoint
Model	010 build	025 build	025 build
B0 train mean	0.000	0.000	0.000
B4 query pair only	0.390	0.390	0.000
B3 screen mean	0.390	0.390	0.142
B1 additive per-gene ridge	0.400	0.400	0.185
B2 additive plus pair ridge	0.406	0.406	0.180
B5 embedding MLP, 3 seeds	0.426 ± 0.001	0.432 ± 0.005	0.141 ± 0.007
CGT, three 010 checkpoints, test	0.438 to 0.455		
CGT GH 1598, val max, epoch 14 of 36		0.446	
CGT GH 1640, val max, epoch 7 of 36			0.199

Arm R is the 010 result. Every closed-form model agrees with its 010 value to 3×10^{-13}, which follows from the record-level label and split identity of Section 2 rather than testing it. B5 differs by 0.006 against a three-seed standard deviation of 0.005; the 025 fit ran on CPU where 010's ran on GPU, and the cause is untraced. GH 1598 reaches a validation maximum of 0.446 at epoch 14, within 0.001 of the lower end of the 0.447 to 0.462 band the three 010 checkpoints recorded on validation.

On the random split the ladder is mostly screen identity. B4 knows only which query pair a triple carries and reaches 0.390, 97 percent of the additive ridge and 86 percent of the best checkpoint. The transformer's margin over B1 is real, +0.038 to +0.055 with paired-bootstrap intervals that exclude zero in the 010 analysis, and it is small.

Holding out screens collapses the ladder and inverts it. On arm Q the additive ridge falls from 0.400 to 0.185 and B4 to zero by construction. B5 falls from 0.432 to 0.141, below the ridge: the capacity that helped on unseen records of seen screens hurts on unseen screens. B3 falls back from a screen mean to a gene mean and lands at 0.142.

One disjoint split is a small sample of screens. Panel d is the reason arm Q's numbers are read as one draw rather than as the disjoint null. Across five disjoint folds on the 010 build the ridge spans 0.088 to 0.151, 0.127 ± 0.033, and arm Q's 0.185 lies above every fold. The same holds for B2, B3 and B5, so arm Q is a comparatively easy draw of 46 held-out screens. A transformer number on arm Q is compared to the arm Q nulls, on the same 46 screens, and not to the fold mean.

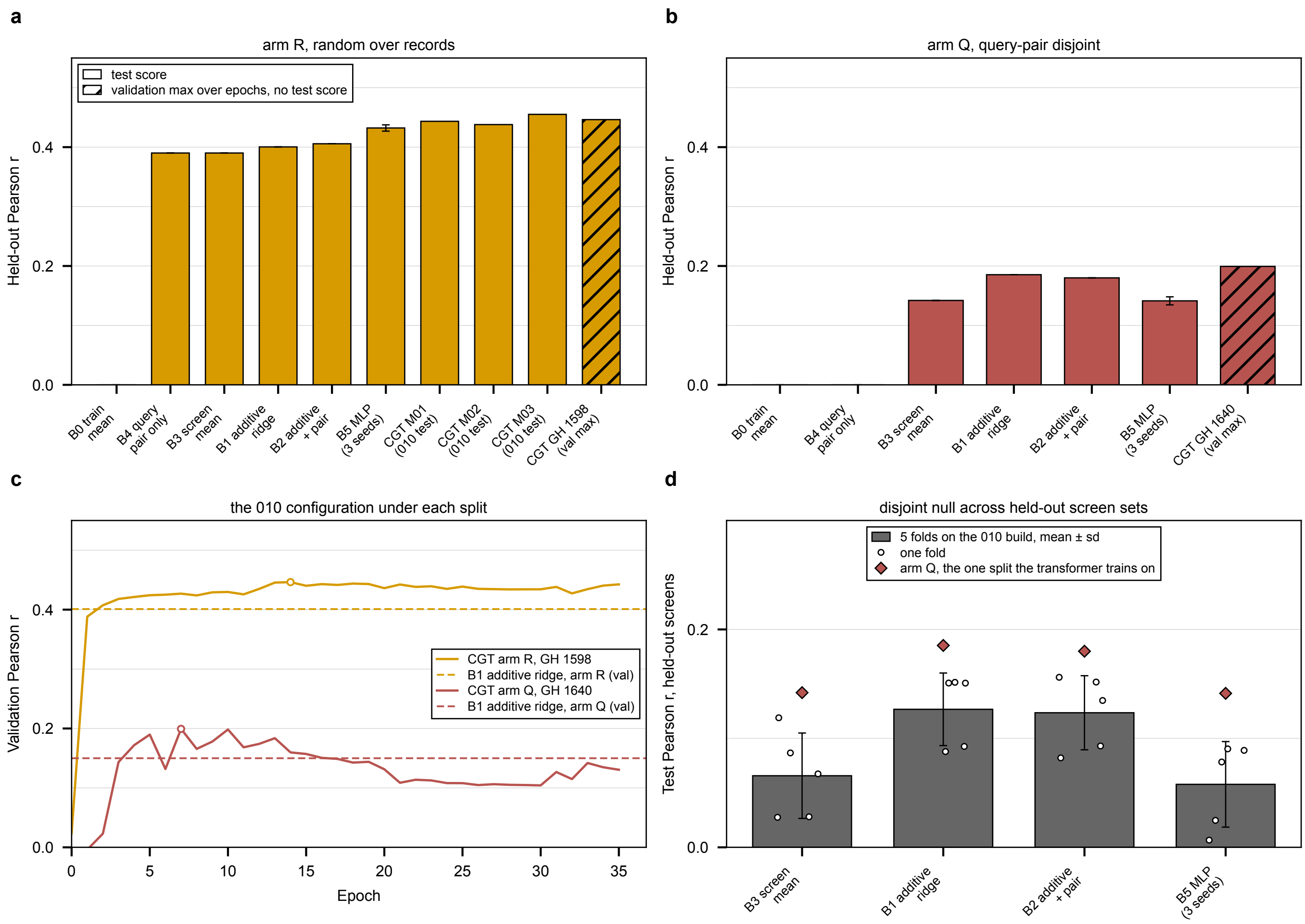

Figure 1: The ladder under both splits. **a**, arm R, random over records: the six baselines on the 37,673 test records, the three 010 checkpoints on the 010 test split, and the 025 replication GH 1598 as a validation maximum, hatched. **b**, arm Q, query-pair disjoint, on the 37,791 test records of 46 held-out screens, with GH 1640 hatched. In both panels only B5 carries an error bar, the standard deviation over its three seeds; every other bar is one deterministic fit. **c**, validation Pearson per epoch for the two arms of the 010 configuration, each against the additive ridge fit on its own validation part; open circles mark each run's best epoch, the value the hatched bars carry. **d**, how far the disjoint null moves with the choice of held-out screens: five disjoint folds on the 010 build per model, mean and standard deviation with each fold as a point, and the arm Q single split as a diamond. B0 and B4 are zero on every fold and are omitted.

5 The transformer on held-out screens ✗

Four transformer runs have trained on arm Q. One is the 010 configuration. The other three come from the joint-fitness-head branch (PR #346, not yet on `main`): they replace the learnable gene table with a fixed sequence embedding of matched parameter count, read out through a perturbed CLS token, and train at a constant learning rate for 30 epochs. They share the split and the graph penalty with GH 1640 and nothing else that a replicate would have to share, so they are four runs on one split, not one run replicated.

Table 5: Every transformer run on arm Q. Epochs is the number logged: GH 1640 was stopped by the 12 hour wall clock partway through epoch 36, and the IGB runs finished their 30. The validation maximum is an upward-biased order statistic over the logged epochs. No run has a test score. Configs of the IGB runs are on branch `feat/025-fitness-joint-head`.

Run	Config	Gene input	Schedule, readout	Epochs	Val max (epoch)	Val last
GH 1640	`cgt_s0_q_kl_004`	learnable table	cosine, 010 schedule, mean	36	0.199 (7)	0.131
IGB 2391132	`cgt_s0_q_kl_emb_017`	four-region sequence composite	constant 2.5×10^{-4}, perturbed CLS	30	0.262 (2)	0.222
IGB 2391133	`cgt_s0_q_kl_calm_020`	CaLM codon embedding	constant 2.5×10^{-4}, perturbed CLS	30	0.252 (7)	0.208
IGB 2391134	`cgt_s0_q_kl_prot_021`	ProtT5 protein embedding	constant 2.5×10^{-4}, perturbed CLS	30	0.270 (3)	0.135

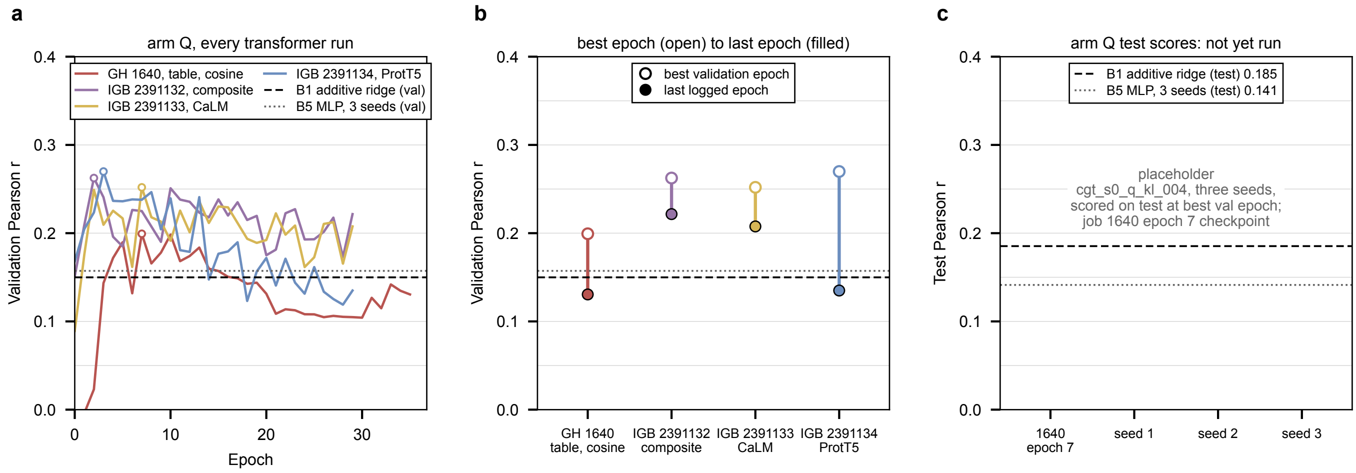

Figure 2: Every transformer run on the query-pair-disjoint split. **a**, validation Pearson per epoch on the 37,705 validation records of 43 held-out screens, with the additive ridge and the three-seed MLP fit on the same validation part as horizontal lines; open circles mark each run's best epoch. **b**, the same runs reduced to their best epoch (open) and last logged epoch (filled). **c**, placeholder for the test comparison that has not been run: the arm Q test nulls are drawn, and the empty slots are the epoch 7 checkpoint of GH 1640 and three seeds of the 010 configuration, each to be scored on the 46 test screens at its best validation epoch.

Every run peaks early and falls. GH 1640 passes the additive validation null of 0.150 in epoch 4, peaks at 0.199 in epoch 7, is last above the null in epoch 16 and ends at 0.131, below every null but B3. The three IGB runs peak higher, 0.252 to 0.270 in epochs 2 to 7, and decline from there. The composite and CaLM embeddings end at 0.222 and 0.208 and stay above the ridge through their last five epochs; ProtT5 ends at 0.135, level with GH 1640. Arm R has the opposite shape, a margin over its null that appears in epoch 2 and holds for the rest of the run (Figure 1c).

What the early peak does and does not say. On validation, a transformer holds a margin over additivity for some epochs on screens it has never seen, and the size of that margin depends on the gene input and the schedule. Two things keep this from being a result. The maxima are order statistics over 30 to 36 noisy epochs, so each is biased upward by an amount the nulls, which are single fits, do not share. And validation is 43 screens; the test part is a different 46, and Figure 1d shows how far one draw of screens moves the null. A checkpoint scored on test at its validation-selected epoch is the number that compares to Table 4, and no run has one.

Hypothesis, untested. The decline after the peak is consistent with the model fitting screen-specific structure of the training pairs that does not transfer, the quantity B4 measures. No measurement here separates that from ordinary overfitting of label noise, and no mechanism is claimed.

6 What is left before this enters the main text ✗

1. **Score GH 1640's epoch 7 checkpoint on the arm Q test part.** The checkpoint survives (`$DATA_ROOT/models/checkpoints/327csnlk-best-pearson-epoch=07-val*`). One scoring pass fills the first slot of Figure 2c and gives the first transformer test number against B1 0.185 and B5 0.141.

2. **Replicate the 010 configuration on arm Q.** Three seeds of `cgt_s0_q_kl_004` with a test evaluation at the best validation epoch, under one fixed epoch budget. GH 1640 peaked at epoch 7 and 36 epochs cost the 12 hour wall clock, so a 20 epoch budget with checkpoint-on-best is enough. These fill the remaining slots of Figure 2c and give the transformer row of Table 4 a spread.

3. **Decide whether the constant-learning-rate variants belong in the ladder.** The IGB runs are a different model, readout and schedule. If they are to be compared to the nulls they need the treatment of item 2: a learnable-table control at constant learning rate on arm Q (`cgt_s0_q_kl_ctrl_016`, which had no run on W&B as of 2026-09-11), seeds, and test scores. Otherwise they stay in Figure 2a,b as validation trajectories and do not enter Figure 1.

4. **A paired bootstrap on arm Q test.** The 010 analysis established the random-split margin with a paired bootstrap over test records. The same 2,000-resample interval on the arm Q predictions is what turns a transformer minus B1 difference into a claim.

5. **Fold spread for the transformer, or an explicit statement that it is unaffordable.** Panel d of Figure 1 is five folds for the nulls. A transformer fold costs about 12 hours of one GilaHyper node per 36 epochs, so five folds by three seeds is 15 such runs. If the main text reports arm Q alone it should say so and cite the fold spread of the null as the scale of the split-to-split variation.

6. **B5 on GPU.** The 025 B5 fit ran on CPU and differs from 010's GPU fit by 0.006 on arm R. A GPU refit closes the one row of Table 4 that is not an exact reproduction.

7. **Bibliography.** The Visani preprint and Kuzmin 2018 are named by DOI in prose. A Zotero collection for this document, then `make bib-pull`, makes them citations.

What is not carried over from the 010 report. The reading of the transformer code that establishes it is a set function over the perturbed gene identities, the graph penalty's share of the loss, the neighbor recall of the regularized heads, the matched pair of runs without the penalty, the record-by-record agreement between models, and the gene-index defect of the inference runs. Each is in `notes-tex/010-additive-baselines` and is cited from there when the main text needs it.